==========

Pre_processing

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```
# Import libraries
# -----------------
import pandas as pd
```

```
# Load Dataset
# ---------------
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving data.csv to data.csv
```

```
df = pd.read_csv("data.csv", encoding='latin1')
```

```
# --------------------
# Data Understanding
# --------------------
```

```
print(df.shape)
```

```
(536641, 8)
```

```
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.0	United Kingdom

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 536641 entries, 0 to 536640
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    536641 non-null  object
 1   StockCode    536641 non-null  object
 2   Description  535187 non-null  object
 3   Quantity     536641 non-null  int64
 4   InvoiceDate  536641 non-null  object
 5   UnitPrice    536641 non-null  float64
 6   CustomerID   401604 non-null  float64
 7   Country      536641 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 32.8+ MB
```

```
df.describe()
```

	Quantity	UnitPrice	CustomerID
count	536641.000000	536641.000000	401604.000000
mean	9.620029	4.632656	15281.160818
std	219.130156	97.233118	1714.006089
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13939.000000
50%	3.000000	2.080000	15145.000000
75%	10.000000	4.130000	16784.000000
max	80995.000000	38970.000000	18287.000000

```
df.isnull().sum()
```

	0
InvoiceNo	0
StockCode	0
Description	1454
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	135037
Country	0

dtype: int64

```
df['Description']
```

	Description
0	WHITE HANGING HEART T-LIGHT HOLDER
1	WHITE METAL LANTERN
2	CREAM CUPID HEARTS COAT HANGER
3	KNITTED UNION FLAG HOT WATER BOTTLE
4	RED WOOLLY HOTTIE WHITE HEART.
...	...
536636	PACK OF 20 SPACEBOY NAPKINS
536637	CHILDREN'S APRON DOLLY GIRL
536638	CHILDRENS CUTLERY DOLLY GIRL
536639	CHILDRENS CUTLERY CIRCUS PARADE
536640	BAKING SET 9 PIECE RETROSPOT

536641 rows × 1 columns

dtype: object

```
# -----------------------
# Handle Missing Values
# -----------------------
```

```
df.dropna(subset=['Description'])
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
...	...	...	...	...	...	...	...	...
536636	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	12/9/2011 12:50	0.85	12680.0	France
536637	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	12/9/2011 12:50	2.10	12680.0	France

```
df.isnull().mean()*100
```

	0
InvoiceNo	0.000000
StockCode	0.000000
Description	0.270945
Quantity	0.000000
InvoiceDate	0.000000
UnitPrice	0.000000
CustomerID	25.163377
Country	0.000000

dtype: float64

```
# ---------------
# Data Cleaning
# ---------------
```

```
df=df.drop_duplicates()
```

```
df=df[df['Quantity']>0]
df=df[df['UnitPrice']>0]
```

```
# ----------------------
# Data Type Conversion
# ----------------------
```

```
df['CustomerID']=df['CustomerID'].astype(str)
```

```
df['InvoiceDate']=pd.to_datetime(df['InvoiceDate'])
```

```
# ---------------------
# Feature Engineering
# ---------------------
```

```
df['TotalPrice']=df['Quantity']*df['UnitPrice']
```

```
# ------------------------
# Final Dataset Overview
# ------------------------
```

```
print ("Final Dataset Shape:",df.shape)
df.head(10)
```

```
Final Dataset Shape: (524878, 9)
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPrice
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850.0	United Kingdom	15.30
6	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	2010-12-01 08:26:00	4.25	17850.0	United Kingdom	25.50
7	536366	22633	HAND WARMER UNION JACK	6	2010-12-01 08:28:00	1.85	17850.0	United Kingdom	11.10
8	536366	22632	HAND WARMER RED POLKA DOT	6	2010-12-01 08:28:00	1.85	17850.0	United Kingdom	11.10
9	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	2010-12-01 08:34:00	1.69	13047.0	United Kingdom	54.08

```
df.tail(10)
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPrice
536631	581587	22726	ALARM CLOCK BAKELIKE GREEN	4	2011-12-09 12:50:00	3.75	12680.0	France	15.00
536632	581587	22730	ALARM CLOCK BAKELIKE IVORY	4	2011-12-09 12:50:00	3.75	12680.0	France	15.00
536633	581587	22367	CHILDRENS APRON SPACEBOY DESIGN	8	2011-12-09 12:50:00	1.95	12680.0	France	15.60
536634	581587	22629	SPACEBOY LUNCH BOX	12	2011-12-09 12:50:00	1.95	12680.0	France	23.40
536635	581587	23256	CHILDRENS CUTLERY SPACEBOY	4	2011-12-09 12:50:00	4.15	12680.0	France	16.60
536636	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	12680.0	France	10.20
536637	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	12680.0	France	12.60

```
df.sample(10)
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPrice
193205	553673	21672	WHITE SPOT RED CERAMIC DRAWER KNOB	12	2011-05-18 12:46:00	1.25	16814.0	United Kingdom	15.00
438081	574656	23103	JINGLE BELL HEART DECORATION	24	2011-11-06 11:36:00	1.65	16434.0	United Kingdom	39.60

```
df.isnull().sum()
```

307567	564150	20828	GLITTER BUTTERFLY CLIPS	12	2011-08-23 12:30:00	2.55	14936.0	Channel Islands	30.60
136850	548191	21656	RIDGED GLASS POSY VASE	1	2011-03-29 15:20:00	3.29	nan	United Kingdom	3.29
	943	22632	HAND WARMER RED RETROSPOT	12	2011-08-31 13:23:00	2.10	17706.0	United Kingdom	25.20
	093	22627	MINT KITCHEN SCALES	2	2011-05-31 13:15:00	8.50	12720.0	Germany	17.00
495130	578696	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-11-25 10:02:00	4.15	13985.0	United Kingdom	16.60
98856	544788	21497	FANCY FONTS BIRTHDAY WRAP	25	2011-02-23 12:18:00	0.42	13089.0	United Kingdom	10.50
	735	21564	PINK HEART SHAPE LOVE BUCKET	1	2011-04-04 10:52:00	2.95	15039.0	United Kingdom	2.95

InvoiceNo	0
StockCode	0
Description	0
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	0
Country	0
TotalPrice	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 524878 entries, 0 to 536640
Data columns (total 9 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    524878 non-null  object
 1   StockCode    524878 non-null  object
 2   Description  524878 non-null  object
 3   Quantity     524878 non-null  int64
 4   InvoiceDate  524878 non-null  datetime64[ns]
 5   UnitPrice    524878 non-null  float64
 6   CustomerID   524878 non-null  object
 7   Country      524878 non-null  object
 8   TotalPrice   524878 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(1), object(5)
memory usage: 40.0+ MB
```

Start coding or generate with AI.